



NOISE SUPPRESSION AND GPS SIGNAL PROCESSING FOR ROCKET LAUNCH TRAJECTORIES

Amin, Riyaan¹ & Marsellos, A.E.²

¹Foothill High School, 4375 Foothill Rd, Pleasanton, CA, USA

²Dept. Geology, Environment and Sustainability, Hofstra University, Hempstead, NY, USA



Abstract:

Global Positioning System (GPS) data play a central role in rocket launch activities, as they provide indispensable information for navigation, trajectory adjustments, and payload delivery. GPS measurements, however, are often corrupted by noise resulting from atmospheric conditions, multipath effects, and instrumentation limitations. This study examines noise suppression and signal enhancement techniques for GPS datasets to increase reliability and accuracy during rocket launch operations.

Using statistical and computational methods, filtering and outlier detection were applied to sampled GPS data, demonstrating effective suppression of spurious signals while preserving true positional trends. The research evaluates several noise suppression schemes, including moving average smoothing, median filtering, and spectral and adaptive techniques, with a focus on balancing signal preservation and error reduction. Through the advancement of GPS signal processing methods, this research contributes to safer and more accurate rocket launches, with broader applications across aerospace engineering, navigation, and autonomous systems.

Background Information:

- GPS is critical for navigation, trajectory corrections, and payload delivery in rocket launches.
- Sources of GPS noise: atmospheric interference, multipath reflections, receiver error
- 04A_Tutorial_R_Intro_to_cleaning
- Noisy data can lead to miscalculated trajectories → higher mission risk.
- Reliable noise reduction is essential for aerospace, autonomous vehicles, and navigation systems.

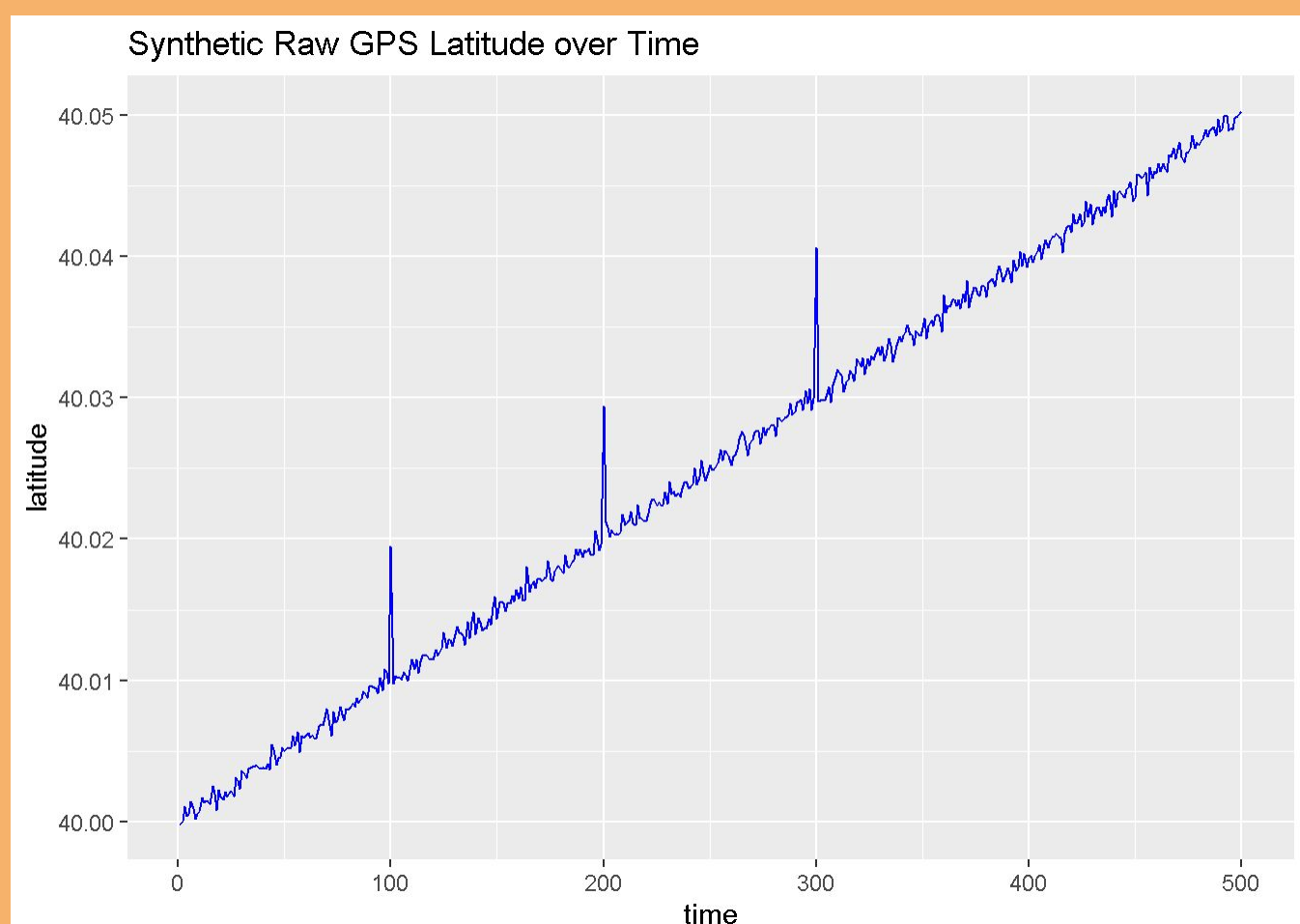


Figure 1: Raw GPS latitude over time. Synthetic dataset shows clear noise and injected outliers (spikes).

Method/Materials:

- Data**
 - Synthetic GPS data with noise and injected outliers.
- Tools:**
 - R
- RStudio (v.4.4.3)**
 - ggplot2, signal, zoo, kza, forecast packages
 - Low-Pass Butterworth filter
 - Z-score outlier detection
 - Moving average smoothing
 - Kolmogorov-Zurbenko Adaptive (KZA) filter
 - Outlier detection + replacement (rolling mean)
 - Time-series based correction (tsoutliers)

Results:

- Figure 1 shows raw GPS data with noise and injected outliers.
- Figure 2 demonstrates the low-pass Butterworth filter, which removes high-frequency noise but risks over-smoothing.
- Figure 4 (Moving Average) shows edge distortion.
- Figure 5 (tsoutliers) best preserves trajectory shape while removing spikes.
- Figure 3 summarizes each method's strengths and weaknesses.
- Main finding: The Kolmogorov-Zurbenko adaptive filter and tsoutliers provided the best balance between smoothing and trend preservation, resulting in improved trajectory stability.

$$G(\omega) = \frac{1}{\sqrt{1 + \omega^{2n}}},$$

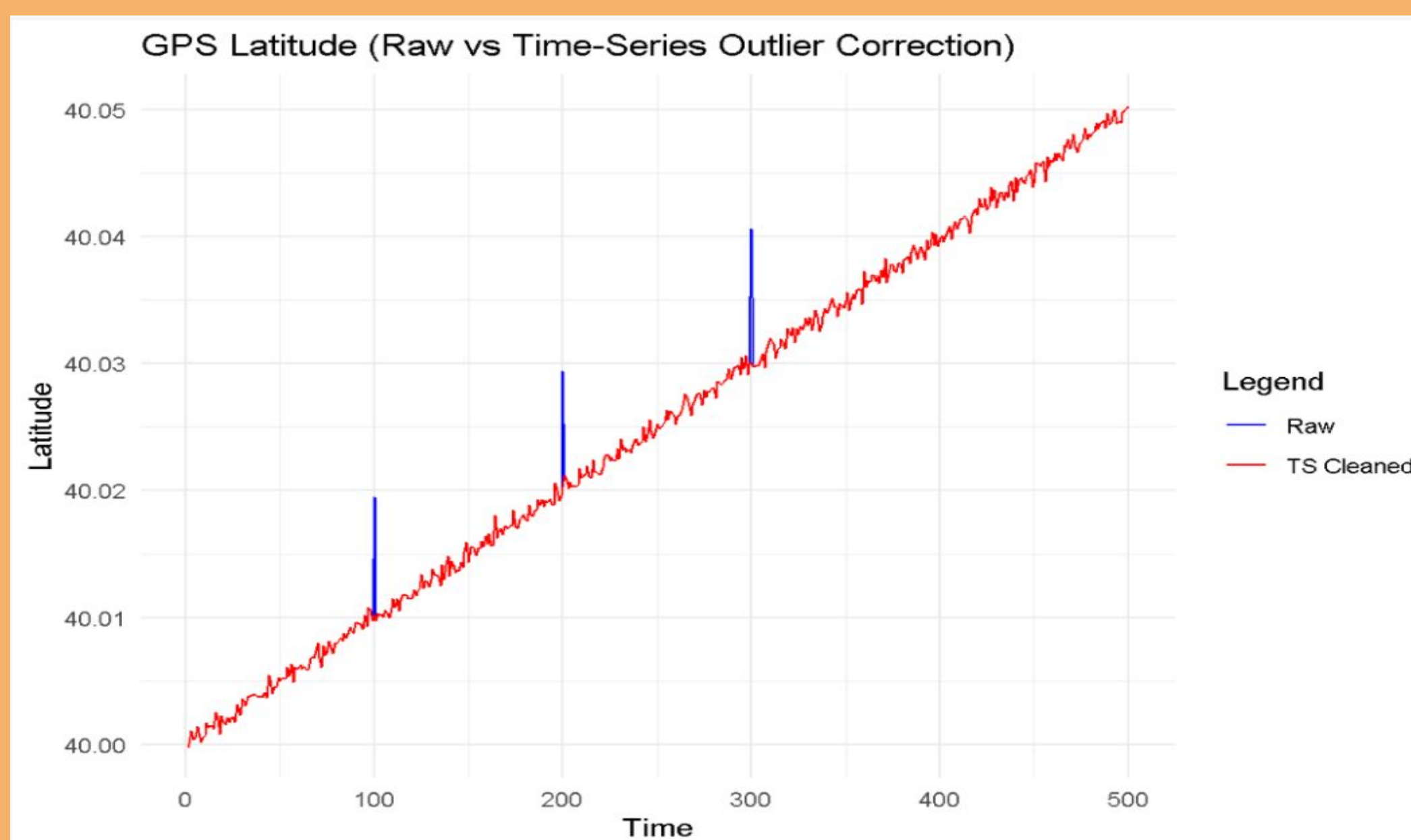


Figure 5: The Raw signal (blue) contains noise spikes that mimic errors during rocket launch tracking. After applying tsoutliers, the cleaned signal (red) preserves the overall trajectory while removing the anomalies.

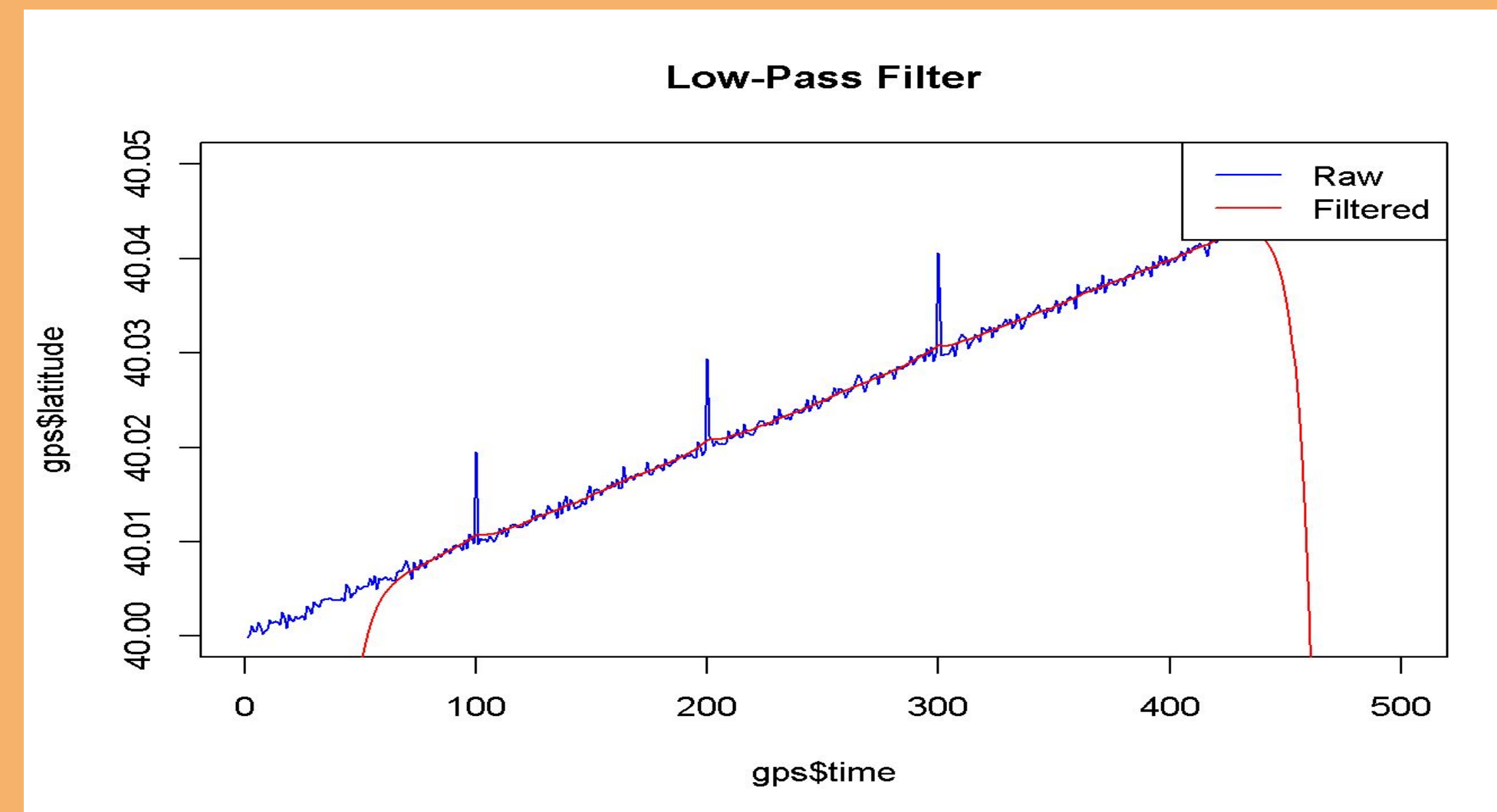


Figure 2: Low-pass Butterworth filter applied to GPS data. Red line indicates smoothed signal; effective at removing high-frequency noise but risks over smoothing true motion.

Method	What it Does	Pros	Cons / Limitations
Low-pass filter (Butterworth)	Removes high-frequency noise	Good for smoothing; simple to apply	Cutoff frequency tricky to choose; can oversmooth
Z-score outlier removal	Removes points far from mean/trend	Easy to understand; good for spikes	Misclassifies if data has strong trend
Moving average	Smooths by averaging neighbors	Simple, intuitive	Loses data at edges; can oversmooth
KZA filter	Adaptive smoothing across dataset	Preserves trends; handles edges better	Smooths entire dataset (may blur fine details)
Outlier detection + replace	Detects & replaces outliers only	Preserves most data; minimal smoothing	Needs good threshold; may miss subtle noise
Time-series (tsoutliers)	Model-based detection (trend-aware)	Accounts for trend/seasonality; robust	Requires time-series structure; more complex to use

Figure 3: Comparison table of all methods. Summarizes strengths, weaknesses, and applicability of each filtering/outlier detection technique for GPS rocket tracking.

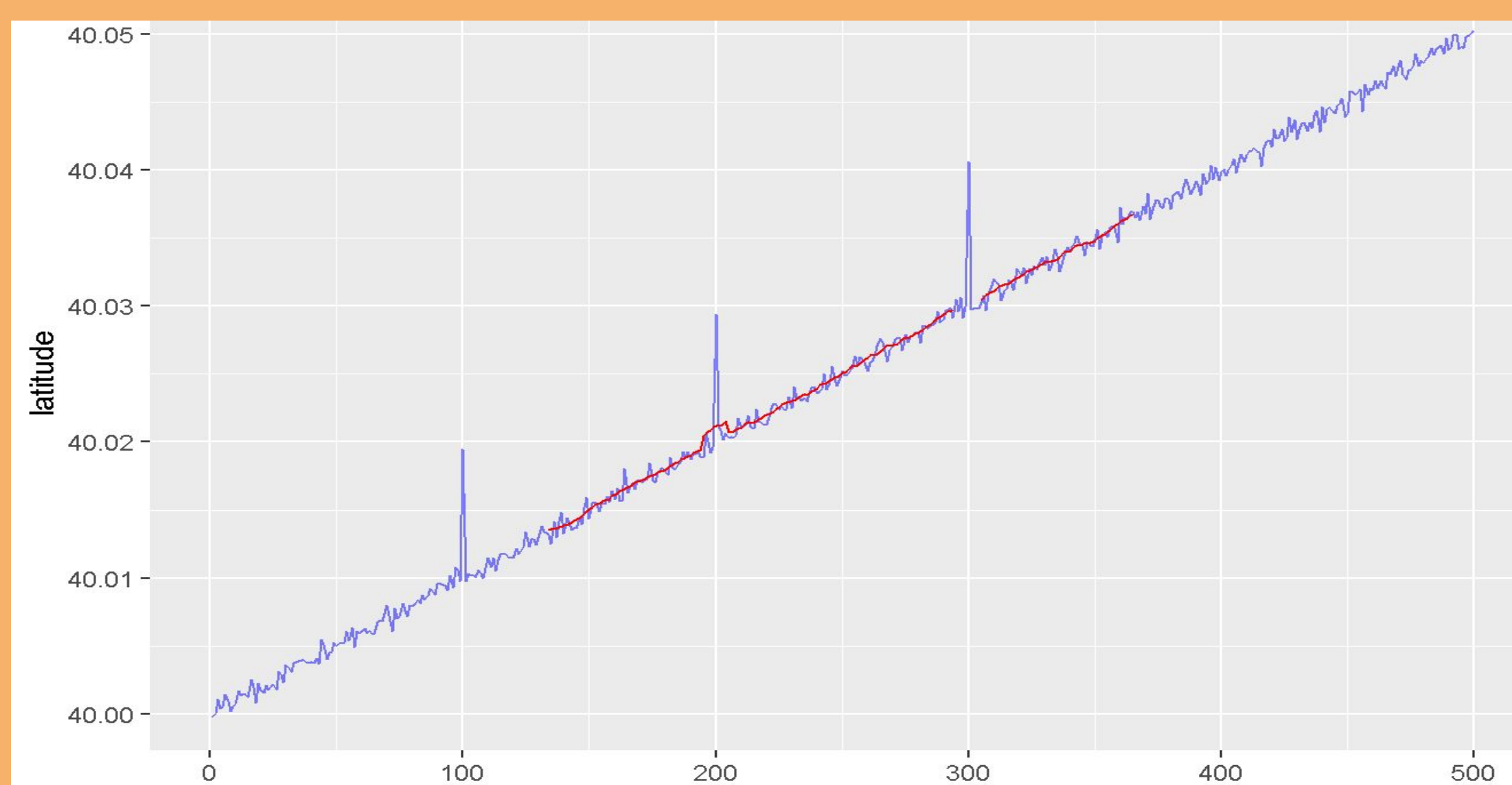


Figure 4: Moving average smoothing after outlier removal. Red line shows smoothed signal; reduces noise but introduces edge effects where data is truncated

Discussion/Conclusion:

- This analysis demonstrates that GPS signal quality can be significantly improved through adaptive filtering and outlier correction. Among all tested methods, the KZ and tsoutliers filters most effectively suppressed noise while maintaining trajectory accuracy.
- These findings support development of more reliable GPS-based navigation for aerospace and autonomous systems.

Acknowledgements:

We would like to thank Dr. Antonios E. Marsellos and the Geology, Environment and Sustainability department at Hofstra University for providing us with the necessary tools to conduct this research.

Credit Authorship Contribution Statement:

Riyaan Amin: Rstudio coding, all figures, tables, poster writing and editing - Marsellos, A.E.: supervision, Rstudio coding, guidance, editing

References:

- R Core Team (2025). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria. <https://www.R-project.org/>.
- Bischi B, Lang M, Bossek J, Horn D, Richter J, Surmann D (2022). *BBmisc: Miscellaneous Helper Functions for B. Bischl*. R package version 1.13, <https://CRAN.R-project.org/package=BBmisc>.
- Chan K, Ripley B (2022). *TSA: Time Series Analysis*. R package version 1.3.1, <https://CRAN.R-project.org/package=TSA>.
- Close B, Zurbenko I, Sun M (2020). *kza: Kolmogorov-Zurbenko Adaptive Filters*. R package version 4.1.0.1, <https://CRAN.R-project.org/package=kza>.
- Desantis, Michael. Callahan's Beach Of Fort Salonga Reopens After Seawall Collapse.
- Garrett Grolemond, Hadley Wickham (2011). Dates and Times Made Easy with lubridate. Journal of Statistical Software, 40(3), 1-25. URL <https://www.jstatsoft.org/v40/i03/>.
- Hester J, Bryan J (2024). *glue: Interpreted String Literals*. R package version 1.7.0, <https://CRAN.R-project.org/package=glue>.
- Tsakiri, K.G. & Marsellos, A.E., 2024. Signal detection in high-noise time series data using R. International conference on Time Series and Forecasting, ITISE2024, Spain, July 2024.
- Moritz S, Bartz-Beielstein T (2017). "imputeTS: Time Series Missing Value Imputation in R." *The R Journal*, 9*(1), 207-218. doi:10.32614/RJ-2017-009 <https://doi.org/10.32614/RJ-2017-009>.
- Ooms J (2024). *curl: A Modern and Flexible Web Client for R*. R package version 5.2.1, <https://CRAN.R-project.org/package=curl>.
- USGS. U.S. Landslide Inventory Web Application. ArcGIS Web Application. March 27, 2024. [U.S. Landslide Inventory \(arcgis.com\)](https://arcgis.com)
- Wickham H, François R, Henry L, Müller K, Vaughan D (2023). *dplyr: A Grammar of Data Manipulation*. R package version 1.1.4, <https://CRAN.R-project.org/package=dplyr>.